

Operating Systems

2520030273

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SEC-5

WEEK-2

2. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved

Code

```
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GNU nano 8.7.1 copy.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int source, destination;
    char buffer[100];
    int n;

    source = open("source.txt", O_RDONLY);

    if (source < 0)
    {
        printf("Cannot open source file\n");
        return 1;
    }

    destination = open("destination.txt",
                      O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination < 0)
    {
        printf("Cannot create destination file\n");
        close(source);
        return 1;
    }

    while ((n = read(source, buffer, sizeof(buffer))) > 0)
    {
        write(destination, buffer, n);
    }

    close(source);
    close(destination);

    printf("File copied successfully\n");

    return 0;
}
```

output

```
kolasani_navadeep@Navadeep-Vitus:~$ nano copy.c
kolasani_navadeep@Navadeep-Vitus:~$ gcc copy.c
kolasani_navadeep@Navadeep-Vitus:~$ ./a.out
Cannot open source file
kolasani_navadeep@Navadeep-Vitus:~$ |
```